

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# PLACE VALUE AND ORDERING INTEGERS, (ORDERING POSITIVE AND NEGATIVE INTEGERS)

**LO: I can order positive and negative integers and place them on a number line.**

## Ordering integers

**Integers** are whole numbers including negatives: ... -3, -2, -1, 0, 1, 2, 3 ... On a number line, numbers **increase to the right**.

**Key rule:** All negative numbers are less than zero. A larger negative magnitude means a smaller value:  $-5 < -2$ .

## Quick examples

|   |           |                                   |
|---|-----------|-----------------------------------|
| ● | $-3 < 2$  | negatives are less than positives |
| ● | $-7 < -2$ | further from zero = smaller       |
| ● | $0 > -4$  | zero is greater than any negative |
| ● | $-5 > -2$ | -5 is further left, so smaller    |
| ● | $-1 < -8$ | -1 is closer to zero, so larger   |

## Key idea

On a number line, left is smaller. For negatives, bigger digit = smaller value.

$$-5 < -3 < 0 < 2 < 5$$

**Further left = smaller value**

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - comparing two integers

Which is larger: -4 or -1?

On number line, -1 is to the right of

-4

Right = larger

$$-1 > -4$$

**-1 is larger**

-1 is closer to zero.

### Example 2 - ordering a set

Put in order: 3, -5, 0, -2, 7.

Smallest to largest: -5, -2, 0, 3, 7

-5 is furthest left

7 is furthest right

**-5, -2, 0, 3, 7**

Place on number line from left to right.

### Example 3 - temperature context

Temperatures:  $-3^{\circ}\text{C}$ ,  $2^{\circ}\text{C}$ ,  $-7^{\circ}\text{C}$ ,  $5^{\circ}\text{C}$ .  
Order from coldest.

Coldest = smallest = -7

Then -3, then 2, then 5

Coldest to warmest: -7, -3, 2, 5

**-7, -3, 2, 5**

Lower temperature = further left on number line.

### Example 4 - finding between

Name an integer between -4 and -1.

Integers between: -3, -2

Both are greater than -4 and less than -1

Either -3 or -2

**-3 or -2**

Between means strictly greater and strictly less.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Comparing pairs

Write < or > between each pair.

- a) -3 \_ 2
- b) -5 \_ -1
- c) 0 \_ -6
- d) -8 \_ -10

### 2. Ordering smallest to largest

Order from smallest to largest.

- a) 4, -2, 0, -5, 3
- b) -1, -8, 6, -3, 2
- c) 10, -10, 0, 5, -5
- d) -4, -2, -7, -1, -9

### 3. Ordering largest to smallest

Order from largest to smallest.

- a) -6, 3, -1, 0, 8
- b) -12, -5, 4, -20, 15
- c) 1, -1, 11, -11, 0
- d) -3, -30, 3, 30, 0

### 4. Temperature problems

Solve each temperature problem.

- a) Coldest: -2, 5, -8, 1, -4
- b) Warmest: -1, -5, 3, -10, 0
- c) How many degrees from -4 to 3?
- d) Temperature drops 5 from -2. New temp?

### 5. Mixed ordering

Solve each.

- a) Integer between -5 and -3
- b) 3 integers less than -2
- c) Smallest: -100, 1, -1, 100, 0
- d) How many integers from -3 to 4 inclusive?
- e)  $|-7|$  compared to  $|3|$

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) -5 is less than -2 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b)  $-3 > 0$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) On a number line, left means smaller ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) -10 is greater than -1 because  $10 > 1$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) Zero is greater than any negative number ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The temperatures in 5 cities are: Moscow **-15°C**, London **4°C**, Oslo **-8°C**, Cairo **18°C**, Reykjavik **-3°C**. Order from coldest to warmest. Find the difference between the coldest and warmest.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A submarine is at **-200 m**. It rises 75 m, then dives 120 m, then rises 50 m. What is its final depth? Express all depths as integers and order them.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

